

4th Semester/ Chemical (Pulp & Paper)
Subject: Mass Transfer

Time : 3 Hrs.

M.M. : 60

SECTION-A

Note: Multiple choice questions. All questions are compulsory (6x1=6)

- Q.1 Which law governs diffusion in mass transfer?
- a) Newton's Law b) Fick's Law
c) Ohm's Law d) Boyle's Law
- Q.2 The term HETP is used for:
- a) Absorption towers b) Extraction columns
c) Packed columns d) Dryers
- Q.3 Desorption is the reverse of :
- a) Condensation b) Absorption
c) Leaching d) Distillation
- Q.4 Dry bulb temperature is:
- a) Measured using wet cloth
b) Same as dew point
c) Air temperature
d) Always zero

- Q.5 Which is a type of adsorption?
 a) Physical b) Electrical
 c) Osmotic d) Ionic
- Q.6 Drying curve shows relation between
 a) Time and temperature
 b) Moisture and Temperature
 c) Moisture and time
 d) Speed and load

SECTION-B

Note: Objective/ Completion type questions. All questions are compulsory. (6x1=6)

- Q.7 What is dew point.
 Q.8 Write the full form of HTU.
 Q.9 What is meant by flooding in absorption columns?
 Q.10 Give an example of adiabatic equipment.
 Q.11 Write the name any two adsorbents.
 Q.12 Drying is mass transfer operation in which _____ is removed from solid.

SECTION-C

Note: Short answer type questions. Attempt any eight questions out of ten questions. (8x4=32)

- Q.13 State and explain Fick's law of diffusion.
 Q.14 Write about any four types of tower packing.
 Q.15 Define relative humidity and percentage humidity.

- Q.16 How does temperature affect the adsorption process?
 Q.17 Draw and explain a typical rate of drying curve.
 Q.18 What is the relation between film and overall mass transfer coefficient?
 Q.19 State any two problems encountered in packed towers and their solutions.
 Q.20 Explain the use of humidity chart with an example.
 Q.21 Give any two differences between physical and chemical adsorption.
 Q.22 What is the difference between bound and unbound moisture?

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x8=16)

- Q.23 Differentiate between absorption and adsorption. What factors should be considered in choosing a solvent for gas absorption?
 Q.24 What are cooling towers? Draw and explain different cooling tower arrangements.
 Q.25 Explain the working of tray dryer with the help of neat diagram.